SPECIAL PROBLEMS IN LINEAR PROGAMMING

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Degeneracy

- When one or more of basic variables are zero.
- If one or more of the basic variables in a basic solution has value zero, the solution is said to be a degenerate basic solution.
- In Simplex, iterations may keep going on without changing the value of objective function. It is called cycling.
- There are methods to remove this problem, but many packages do not incorporate that feature.

Find solution using Simplex method (BigM method)
MAX Z = 2x1 + 6x2
subject to
x1 + 4x2 <= 12
x1 + 2x2 <= 6
and x1,x2 >= 0

Solution:
Problem is

Max $Z = 2x_1 + 6x_2$

subject to

$x_1 + 4x_2 \leq 12$

$x_1 + 2x_2 \leq 6$

and $x_1, x_2 \geq 0;$

The problem is converted to canonical form by adding slack, surplus and artificial variables as appropiate

1. As the constraint-1 is of type ' $\leq$ ' we should add slack variable S_1

2. As the constraint-2 is of type ' $\leq$ ' we should add slack variable S_2

After introducing slack variables

Max $Z = 2x_1 + 6x_2 + 0S_1 + 0S_2$

subject to

$x_1 + 4x_2 + S_1 = 12$

$x_1 + 2x_2 + S_2 = 6$

and $x_1, x_2, S_1, S_2 \geq 0$

Iteration-1		C_j	2	6	0	0	
B	C_B	X_B	x_1	x_2	S_1	S_2	MinRatio $\frac{X_B}{x_2}$
S_1	0	12	1	(4)	1	0	$\frac{12}{4} = 3 \rightarrow$
S_2	0	6	1	2	0	1	$\frac{6}{2} = 3$
$Z = 0$		Z_j	0	0	0	0	
		$Z_j - C_j$	-2	-6 ↑	0	0	

Negative minimum $Z_j - C_j$ is -6 and its column index is 2. So, the entering variable is x_2.

Minimum ratio is 3 and its row index is 1. So, the leaving basis variable is S_1.

$\therefore$ The pivot element is 4.

Entering $= x_2$, Departing $= S_1$, Key Element $= 4$

✚ $R_1(\text{new}) = R_1(\text{old}) \div 4$

✚ $R_2(\text{new}) = R_2(\text{old}) - 2R_1(\text{new})$

Iteration-2		C_j	2	6	0	0	
B	**C_B**	**X_B**	**x_1**	**x_2**	**S_1**	**S_2**	**MinRatio $\frac{X_B}{x_1}$**
x_2	6	3	0.25	1	0.25	0	$\frac{3}{0.25} = 12$
S_2	0	0	**(0.5)**	0	-0.5	1	$\frac{0}{0.5} = 0 \rightarrow$
$Z = 18$		**Z_j**	**1.5**	**6**	**1.5**	**0**	
		$Z_j - C_j$	-0.5 ↑	0	1.5	0	

Negative minimum $Z_j - C_j$ is -0.5 and its column index is 1. So, the entering variable is x_1.

Minimum ratio is 0 and its row index is 2. So, the leaving basis variable is S_2.

∴ The pivot element is 0.5.

Entering $= x_1$, Departing $= S_2$, Key Element $= 0.5$

✚ $R_2(\text{new}) = R_2(\text{old}) \div 0.5$

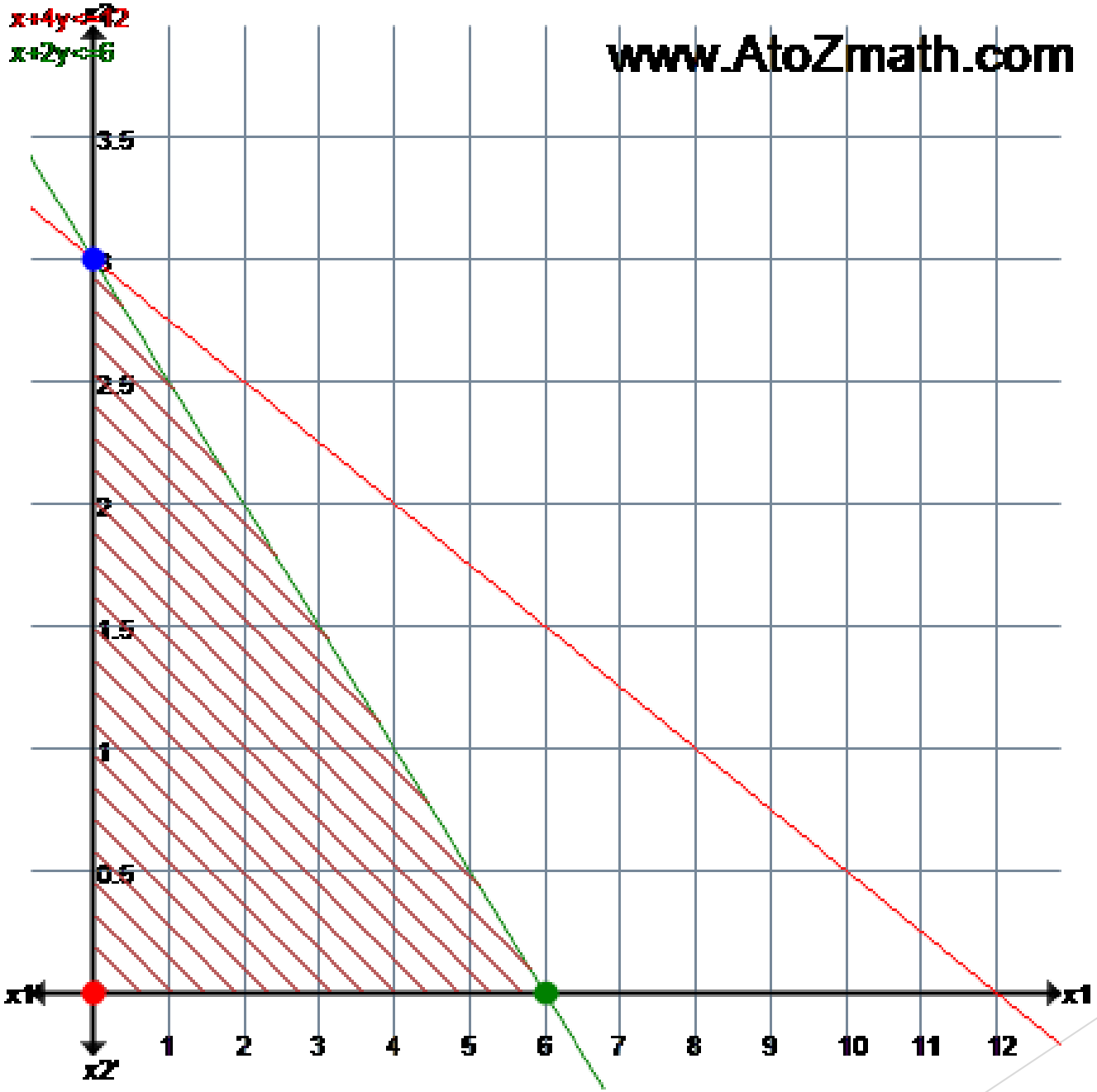
✚ $R_1(\text{new}) = R_1(\text{old}) - 0.25R_2(\text{new})$

Iteration-3		C_j	2	6	0	0	
B	C_B	X_B	x_1	x_2	S_1	S_2	MinRatio
x_2	6	3	0	1	0.5	-0.5	
x_1	2	0	1	0	-1	2	
$Z = 18$		Z_j	2	6	1	1	
		$Z_j - C_j$	0	0	1	1	

Since all $Z_j - C_j \geq 0$

Hence, optimal solution is arrived with value of variables as :
$x_1 = 0, x_2 = 3$

Max $Z = 18$


x+4y<=12
x+2y<=6
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3.5
3
2.5
2
1.5
1
0.5
x1
x2
1
2
3
4
5
6
7
8
9
10
11
12

A case of alternative optima or infinite solutions

Objective:

Maximize

$$Z = x_1 + 2x_2$$

Subject to constraints:

1. $2x_1 + 4x_2 \leq 8$
2. $2x_2 \leq 10$
3. $x_1 \geq 0,\ x_2 \geq 0$ (non-negativity)

$$2x_1 + 4x_2 + s_1 = 8$$
$$2x_2 + s_2 = 10$$
$$x_1, x_2, s_1, s_2 \geq 0$$

Initial Simplex Tableau

Basis	x_1	x_2	s_1	s_2	RHS
s_1	2	4	1	0	8
s_2	0	2	0	1	10
Z	-1	-2	0	0	0

New Tableau After Pivot

Basis	x_1	x_2	s_1	s_2	RHS
x_2	0.5	1	0.25	0	2
s_2	-1	0	-0.25	1	6
Z	0	0	0.5	0	4

Alternate Optimal Tableau:

Basis	x_1	x_2	s_1	s_2	RHS
x_1	1	2	0.5	0	4
s_2	0	2	0.25	1	10
Z	0	0	0.5	0	4

Solution	x_1	x_2	Z
A	0	2	4
B	4	0	4

And **infinitely many** optimal solutions along the line segment connecting A and B:

$$(x_1, x_2) = (\lambda \cdot 4,\ (1-\lambda) \cdot 2), \quad 0 \leq \lambda \leq 1$$

Unbounded Solution

- Maximize $Z = 15A + 20B - MR_1 - MR_2$
- subject to
- $A + 2B - C + R_1 = 10$
- $2A - 3B + D = 6$
- $A + B - E + R_2 = 6$
- $A, B, C, D, E, R_1, R_2 \geq 0$
- Introduce a large penalty M=100 for the artificial variables.

Basis	A	B	C	D	E	R_1	R_2	Solution
Z	−15	−20	0	0	0	100	100	0
R_1	1	2	−1	0	0	1	0	10
D	2	−3	0	1	0	0	0	6
R_2	1	1	0	0	−1	0	1	6

Basis	A	B	C	D	E	R_1	R_2	Solution
Z	−215	−320	100	0	−1	0	0	−1600
R_1	1	2	−1	0	0	1	0	10
D	2	−3	0	1	0	0	0	6
R_2	1	1	0	0	−1	0	1	6

Basis	A	B	C	D	E	R_1	R_2	Solution
Z	−55	0	−60	0	100	160	0	0
B	1/2	1	−1/2	0	0	1/2	0	5
D	7/2	0	−3/2	1	0	3/2	0	21
R_2	1/2	0	1/2	0	−1	−1/2	1	1

It is unbounded.

Infeasible Solution

- Artificial variables do not become zero.
- Show how the M-method will conclude that the following problem has no feasible solution: Maximize z = 2A+5B

Subject to $3A+2B \geq 6$; $2A+B \leq 2$; $A, B \geq 0$.

Maximize $Z = 2A + 5B - MR$

subject to

$3A + 2B - C + R = 6$

$2A + B + D = 2$

$A, B, C, D, R \geq 0$

Introduce a large penalty M=100 for the artificial variables.

Basis	A	B	C	D	R	Solution
Z	−2	−5	0	0	100	0
R	3	2	-1	0	1	6
D	2	1	0	1	0	2

Basis	A	B	C	D	R	Solution
Z	−302	−205	100	0	0	−600
R	3	2	−1	0	1	6
D	2	1	0	1	0	2

Basis	A	B	C	D	R	Solution
Z	0	−54	100	151	0	−298
R	0	1/2	−1	−3/2	1	3
A	1	1/2	0	1/2	0	1

Basis	A	B	C	D	R	Solution
Z	108	0	100	205	0	−190
R	−1	0	−1	−2	1	2
B	2	1	0	1	0	2

Maximize z = 2A+5B
Subject to 3A+2B≥6; 2A+B ≤ 2 ; A, B≥0.

